

Programming Project 2 Question Ideas

1) If you used different arrays/structures/classes to represent the different types of sets, would it be possible to have overloaded methods or operations that would provide the correct functionality regardless of whether or not you were using sets or multi sets? Why or why not? (You do not need to code this, just answer the question).

I think that it would be possible to have overloading methods that would be able to provide correct functionality regardless of whether it is a set or multiset being imputed. If you had maps doing all your multiset operations with lists doing all your set operations it would be pretty easy to just go based off of the input and choose how to do the operation accordingly

2) Would it be possible to use the same data set/structure/class to store sets and multi sets Why or why not?

I think that you could use the same structure and just put everything in a list but I think that would be more confusing. If you had the overridden method from above you could just have a list for sets and maps for multisets. This way you would have to call two different methods for multisets and sets and this would be more annoying to track.

3) How easy or difficult is it to determine the type of set that you need to use based on the users query? Why?(Remember to also consider fuzzy sets in your answer to the following questions.)

I think that it would be easier to determine if a user is trying to use a set or a multi set. If the type of set described was one where all the entries need to be unique you would say it is a set. But if the set being described had the chance or intention of having multiple entries that would be the same then it would be a multi set. I think that it might be a little more confusing to describe a fuzzy set because a user wouldn't describe that it needs degrees of membership and that some of the entries will only be partially associated with the set. They would probably use terms that would be more subtle like partially true or somewhat relevant which could be hard to pick up on.

4) Is it possible to store the data from one type of set (plain sets or multi sets in another type? Would you need to lose data in order to do so? Why?

It is not possible to store multi sets in sets, but I do think that it would be fine to store sets in multisets. If you were to try to store multi sets in a set data structure you would run into the problem of losing all of the duplicates which is a significant amount of data to lose if that isn't your goal to get rid of it. On the other hand if you were to store sets in a multiset nothing would be changed so it would be fine to transfer this way.

5) Discuss what implications your answers to questions 1 – 4 have for someone

trying to code an interface which would allow users to type in natural language queries. When making this project it would be able to hopefully identify sets and multisets pretty well but it might struggle identifying fuzzy sets or it would take more development time to try and make fuzzy sets work well. We could just have overloaded methods that would be able to handle sets and multisets. We would be able to transfer sets to multisets with no consequences but we wouldn't be able to transfer multisets to sets flawlessly because data will be lost. This could partially be solved by making a copy of the multiset and saving it for the user and informing the user that not all of their data will be saved when transferring over to a standard set.

6) Discuss what implications your answers to questions 1 – 5 have for someone trying to code an interface which would allow users to access arbitrary types of data using natural language.

I think that it would take a lot of work trying to make a program that would understand the intent of the user when trying to use certain data types. I think that it might be a little easier when understanding if the user wants a primitive data type like int, boolean, char, etc because you could look for things like numbers being stored or characters. I think the harder thing would be making something like an object like an employee that holds an employee id, a name, and other attached types. It would be hard to make sure that everything is according to the query made by the user.